Guidelines and Evaluation of Clinical Explainable AI in Medical Image Analysis

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Research Problem How can developers choose an AI explainability technique that is optimal for a target clinical task?

Our Solution

Clinical Explainable AI Guidelines

No technical knowledge is required to understand the explanation

Explanation is relevant to clinical decision-making

Explanation should truthfully reflect model decision process

Human judgment on explanation plausibility may reveal decision quality

Computational speed is within clinical users' tolerable waiting time

Explainable AI algorithms

Guideline 1 Understandable — **Guideline 2 Clinical relevant** — **Guideline 3 Truthful** — **Guideline 4 Informative plausibility** — **Guideline 5 Fast** — **Suitable for clinical use**

Evaluation results on 16 heatmap algorithms

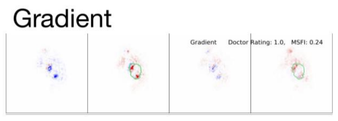
- **G1** Passed
- **G2** Partially passed
- **G3** Not passed
- **G4** Not passed
- **G5** Mostly passed
- STOP The evaluated heatmap methods did not meet G3 and G4, thus cannot be recommended for clinical use.

G1 Understandability & G2 Clinical relevance

Systematic evaluation on 16 heatmap algorithms

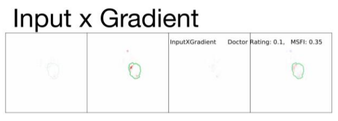
Gradient-based algorithms

Gradient



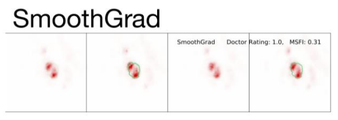
Guided Grad-CAM

Input x Gradient



Integrated Gradient

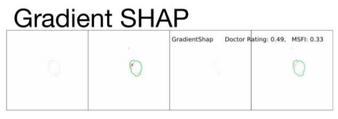
SmoothGrad



DeepLIFT

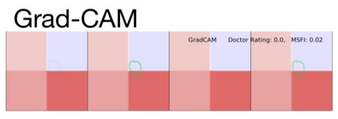
De-convolution

Gradient SHAP



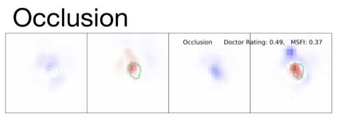
Guided Backpropagation

Grad-CAM



Perturbation-based algorithms

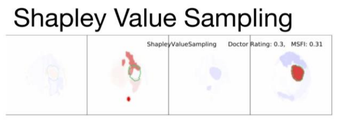
Occlusion



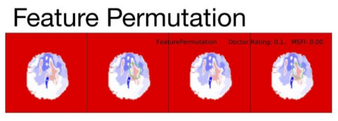
LIME

Feature Ablation

Shapley Value Sampling



Feature Permutation

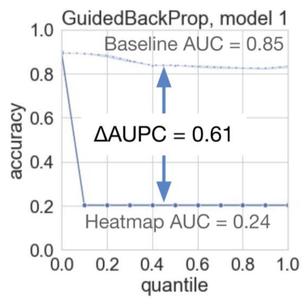


Kernel SHAP

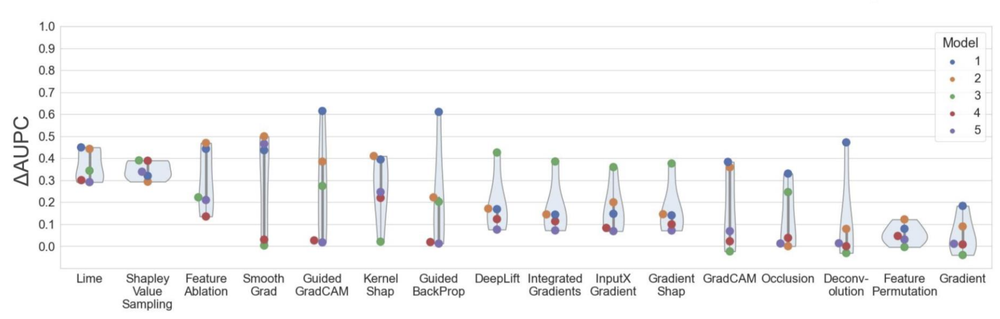
G3 Truthfulness

1. Gradual feature removal - feature level — **All algorithms failed**

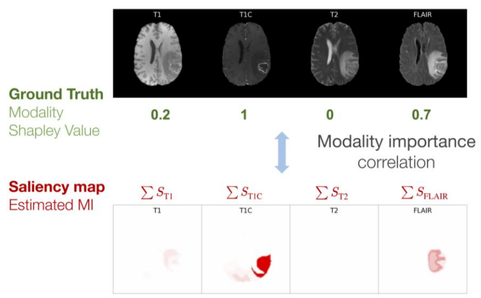
- Low ΔAUPC score
- Not stable across similarly-trained models

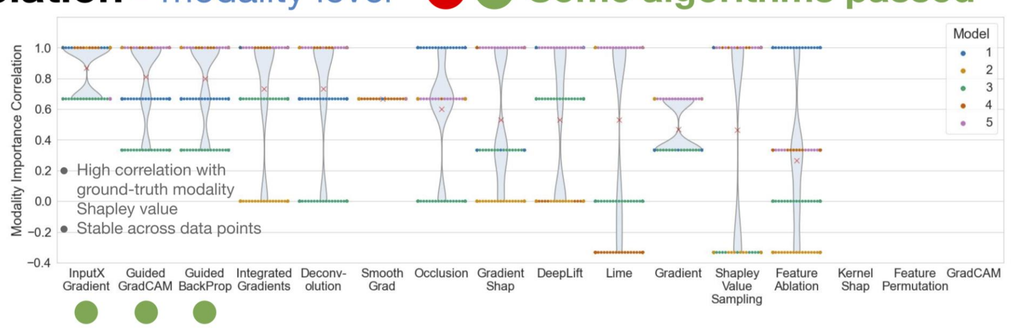


- Gradually remove heatmap features from the most to the least important ones in input
- Plot model performance change w.r.t each removal step
- Calculate the area under the curve difference (ΔAUPC) between an algorithm and its baseline



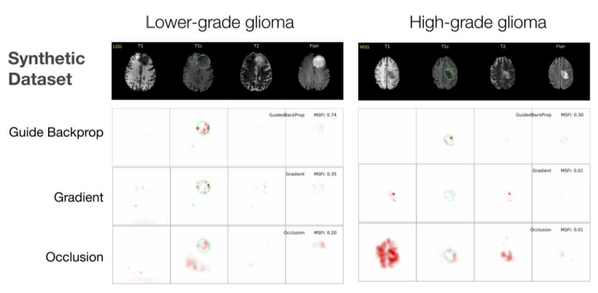
2. Modality importance correlation - modality level — **Some algorithms passed**

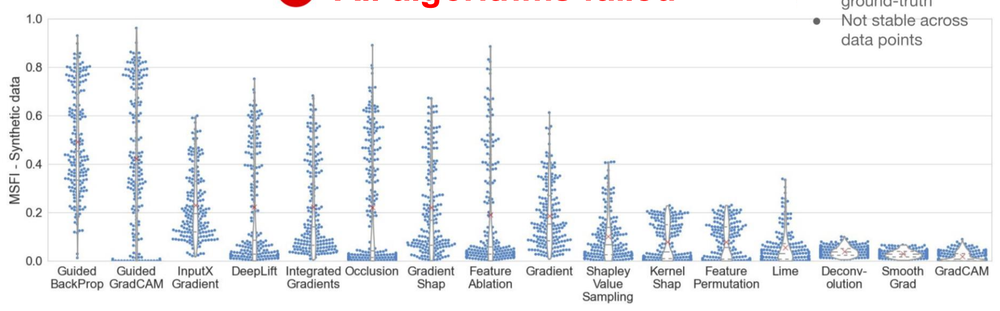




3. Synthetic data experiment - feature level — **All algorithms failed**

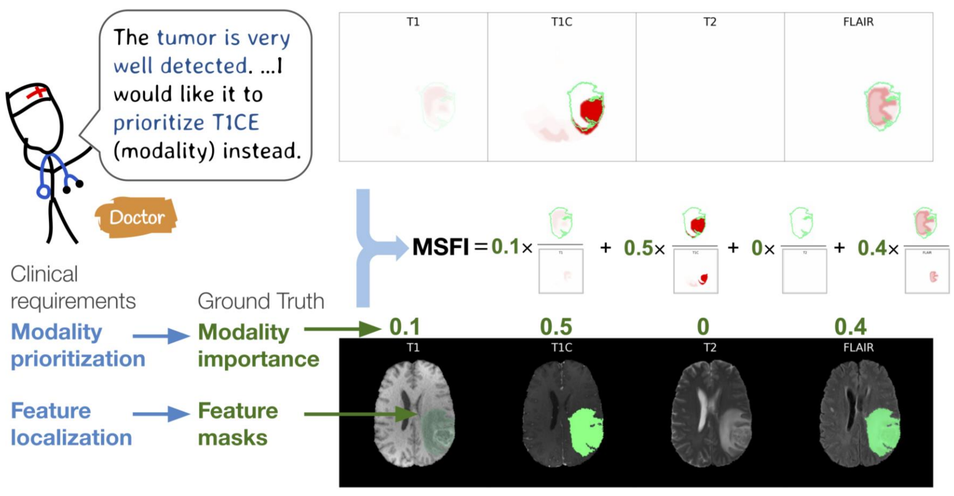
- Low agreement with ground-truth
- Not stable across data points



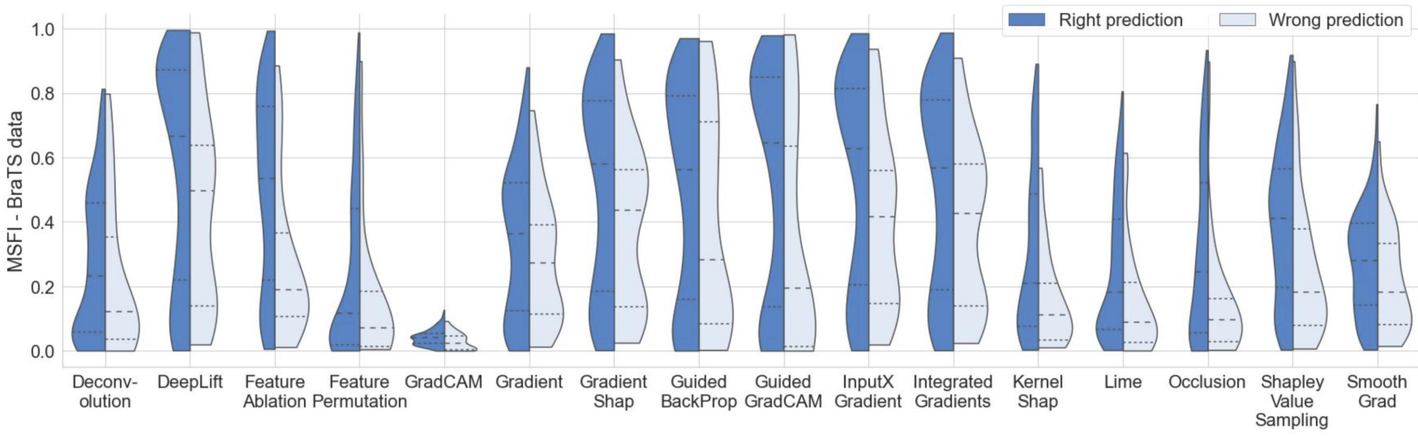


G4 Informative plausibility

Step 1. Plausibility quantification



Step 2. Test for plausibility relation with prediction correctness



All algorithms failed

Their plausibilities were not indicative of model decision quality ($p > 0.05$)

- Rethinking explanation goals: explainability problem ≠ localization problem
- Risks of optimizing explanation for plausibility only:
  - a plausible but unfaithful explanation may learn to deceive, rather than help users

Project Webpage
weina.me/clinical_xai_guideline

Project Code
github.com/weinajin/multimodal_explanation